04.1_time_series_model

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```
library(lubridate)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)

all_sales_2019 <- read_csv("data/processed/all_sales_2019_v2.csv")
```

```
## Rows: 185950 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (5): Product, Street, City, ZIP, Month
## dbl  (5): OrderID, QuantityOrdered, PriceEach, TotalSales, Hour
## dttm (1): OrderDateTime
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Aggregate sales by day

```r
daily_sales <- all_sales_2019 %>%
  mutate(Date = as.Date(OrderDateTime)) %>% # Date format
  group_by(Date) %>%
  summarize(TotalSales = sum(TotalSales)) %>%
  arrange(Date) # data is ordered chronologically
```

create a Time Series Object

```r
# use frequency = 7 >>> weekly seasonality
sales_ts <- ts(daily_sales$TotalSales, frequency = 7)
```

check for adf test

```r
print(adf.test(sales_ts))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  sales_ts
## Dickey-Fuller = -2.3084, Lag order = 7, p-value = 0.4468
## alternative hypothesis: stationary
```

Fit the ARIMA Model

```r
arima_model <- auto.arima(sales_ts, seasonal = TRUE)
```

print Model Summary

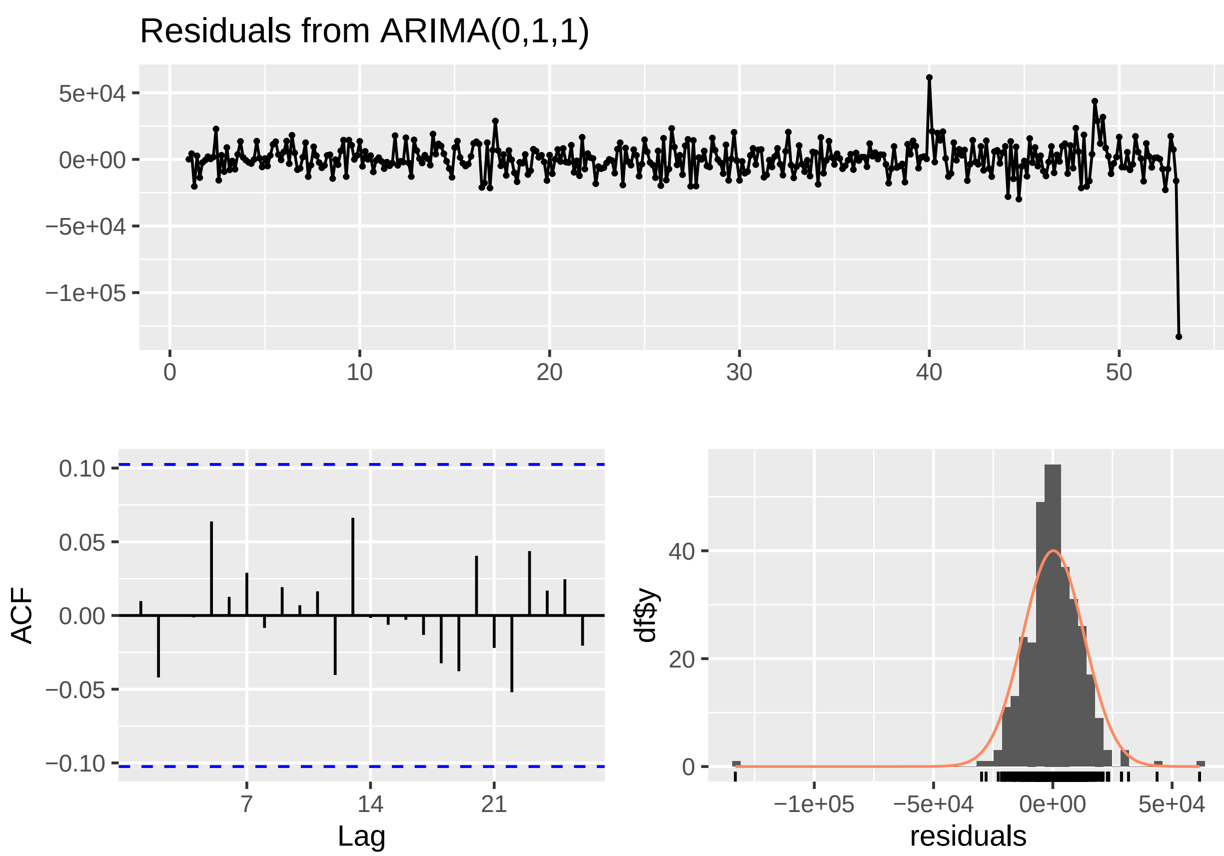
```r
print(summary(arima_model))
```

```
## Series: sales_ts
## ARIMA(0,1,1)
##
## Coefficients:
##           ma1
##       -0.6378
## s.e.   0.0466
##
## sigma^2 = 167059103:  log likelihood = -3973.1
## AIC=7950.2   AICc=7950.24   BIC=7958
##
```

```
## Training set error measures:
##                    ME     RMSE      MAE       MPE     MAPE      MASE       ACF1
## Training set 215.8196 12889.77 8523.167 -4.734123 13.11872 0.7030069 0.00980849
```

check residuals

```r
checkresiduals(arima_model)
```



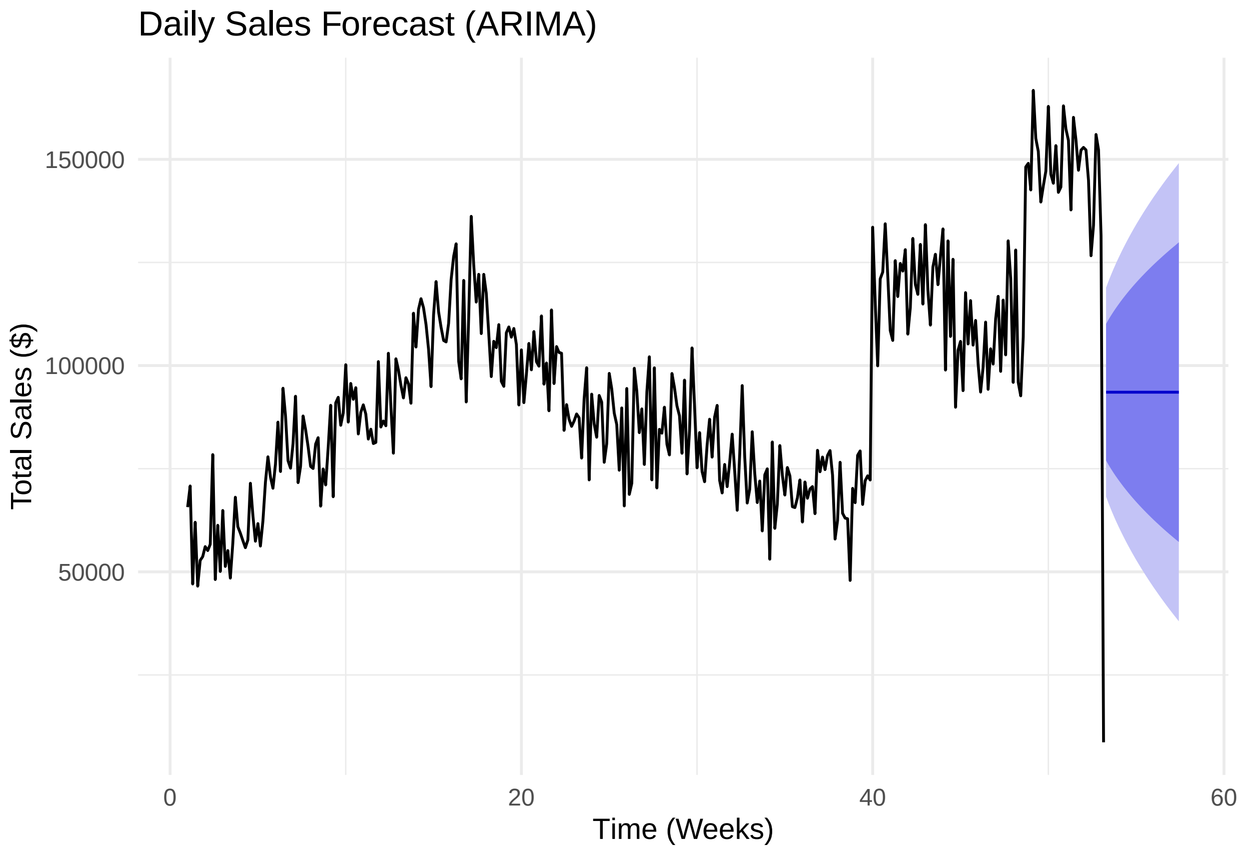

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1)
## Q* = 5.1624, df = 13, p-value = 0.9714
##
## Model df: 1.   Total lags used: 14
```

forecast future sales (Next 30 Days)

```r
sales_forecast <- forecast(arima_model, h = 30)
```

visualization

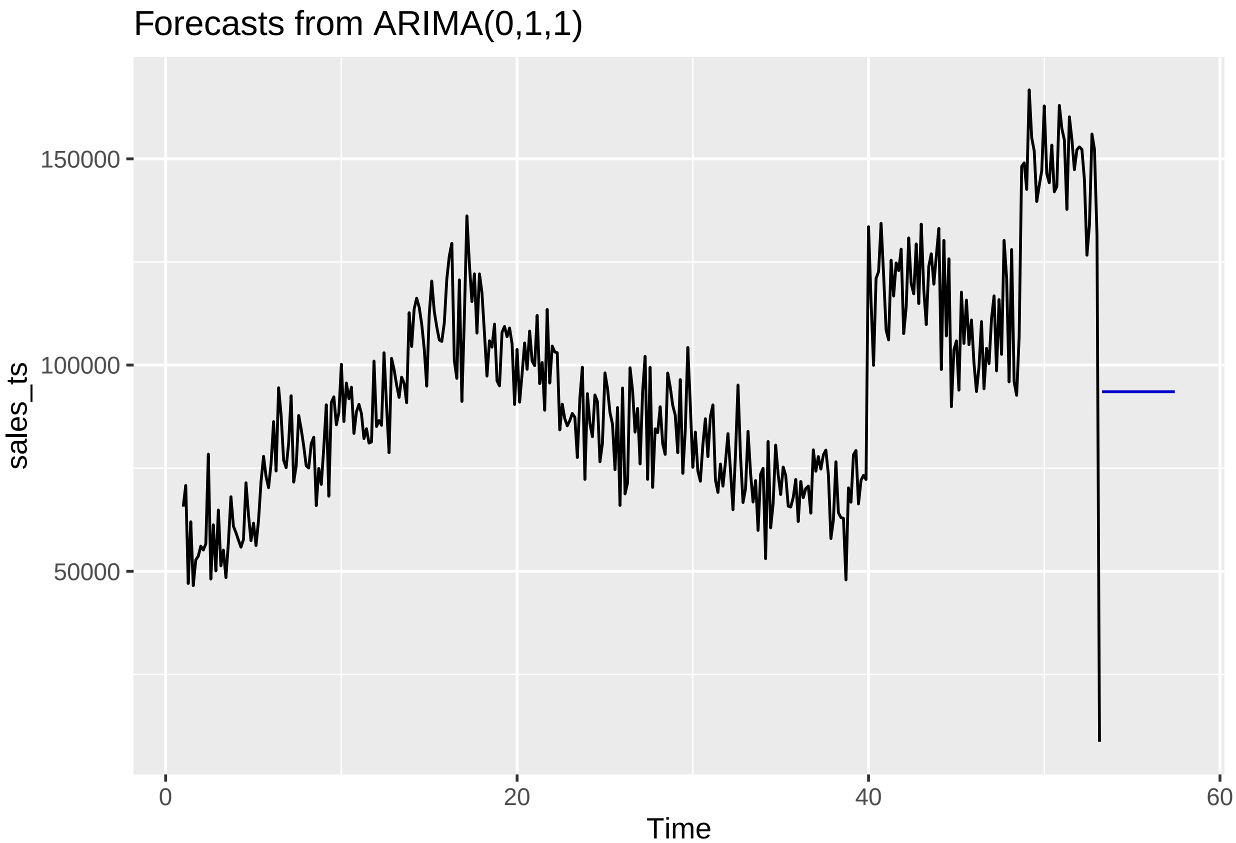
```r
autoplot(sales_forecast) +
  labs(title = "Daily Sales Forecast (ARIMA)",
       x = "Time (Weeks)",
       y = "Total Sales ($)") +
  theme_minimal()
```



visualization without the purple shade

```r
autoplot(sales_forecast, PI = FALSE) +
  geom_line(color = "red", size = 1.5)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Forecasts from ARIMA(0,1,1)
150000
100000
50000
sales_ts
0
20
40
60
Time